

## 2024, Mar. 2024 Release LOC:Acute Adult

## Gallbladder Disorders

Utilization Benchmarks: Multiple options available, see table below for options

## Overview

## Select Day

Initial review (1, 2, 3, 4)

Episode Day 1 (1, 3, 4)

Episode Day 2 (4, 16, 17)

Episode Day 3 (4, 28, 29)

Episode Day 4

Discharge Screens (30)

Utilization Benchmarks

Notes

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## Overview

**Instruction:** This subset is for the medical management of a patient with acute cholecystitis or suspected or confirmed choledocholithiasis. For patients with ascending cholangitis or biliary sepsis, use the Infection: GI/GYN subset. For patients requiring open cholecystectomy, refer to the General Surgical subset. Patients who require laparoscopic cholecystectomy should be managed on an outpatient basis.

## Introduction:

**Acute cholecystitis:** Acute cholecystitis is an inflammation of the gallbladder caused by a transient or permanent obstruction of the cystic duct, usually by gallstones. There are two types of cholecystitis: calculous or acalculous. In calculous, the initial symptoms are epigastric pain, nausea and/or vomiting, and a positive Murphy's sign. Other non-specific symptoms may include chills, malaise, anorexia, low- to moderate-grade fever, tachycardia, and right upper quadrant tenderness. Up to 25%(0.25) of patients will have a palpable gallbladder. Patients with acalculous cholecystitis have similar symptoms but related to dysfunctional emptying of the gallbladder and not due to stones. This form is usually related to trauma or burns, post-surgery and could be related to certain infections.

**Choledocholithiasis:** Choledocholithiasis is the presence of bile stones in the common bile duct (CBD). Most often, CBD stones migrate from the gallbladder into the CBD causing obstruction resulting in right upper quadrant abdominal pain, nausea, and vomiting. Depending on the location of the bile stone, jaundice or pancreatitis may also occur.

## Evaluation and Treatment:

**Acute cholecystitis:** The diagnosis of acute cholecystitis is supported by abdominal ultrasound. Other imaging studies such as computed tomography or magnetic resonance imaging may be performed when the diagnosis remains ambiguous.

Supportive care is the goal of initial treatment in patients with acute cholecystitis. Specifically, care should be directed towards fluid resuscitation, pain management, and broad-spectrum anti-infective therapy. Cholecystectomy is the standard of care for acute calculous cholecystitis. Typically, this can be accomplished using a laparoscopic approach, but an open procedure may be required, particularly when inflammation is severe or other circumstances exist that place the patient at risk for biliary injury using the laparoscopic approach. Laparoscopic cholecystectomy should ideally be performed within 72 hours. Early surgery is associated with better results in comparison to delayed surgery (Gomes et al., World J Gastrointest Surg 2017, 9: 118-26). For those patients who are unable to tolerate urgent surgery due to toxicity or comorbidities, a percutaneous transhepatic or transperitoneal cholecystostomy is the preferred treatment (Knab et al., Surg Clin North Am 2014, 94: 455-70).

**Choledocholithiasis:** Initial assessment for an individual with suspected choledocholithiasis includes physical

examination, laboratory testing, and imaging findings (e.g., ultrasound, computed tomography [CT] scan). Endoscopic retrograde cholangiopancreatography (ERCP) is the gold standard for treatment of choledocholithiasis; however, adverse events can occur with this procedure in 6%(0.06) to 15%(0.15) of individuals and include pancreatitis, infection, bile duct injury, and hemorrhage (Narula et al., Surg Endosc 2020, 34: 1482-91; Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15). Magnetic resonance cholangiopancreatography (MRCP), endoscopic ultrasound, and intraoperative cholangiogram have high sensitivities and specificities for the diagnosis of choledocholithiasis and may prevent injury to the CBD if the diagnosis of choledocholithiasis is indeterminate by other imaging studies; but, if a CBD stone is identified by these modalities, the patient may require a subsequent procedure for stone extraction (Narula et al., Surg Endosc 2020, 34: 1482-91; Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15; Hope et al., Surg Endosc 2017, 31: 2007-16). Treatment recommendations to proceed directly to stone extraction or to perform further diagnostics are based on predictive risk factors that weigh the likelihood of an individual actually having choledocholithiasis against the risk associated with ERCP and stone extraction (Narula et al., Surg Endosc 2020, 34: 1482-91; Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15; Manes et al., Endoscopy 2019, 51: 472-91).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

**Select Day, One:**● **Initial review, One:** <sup>(1, 2, 3, 4)</sup>

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** <sup>(5)</sup>● Acute cholecystitis and, **Both:** <sup>(6)</sup>● Confirmed by imaging, ≥ **One:**

– Ultrasound (US)

– CT

– MRI

– Hepatobiliary scintigraphy

– Unresponsive to at least 2 doses of analgesics

● **ACUTE, One:** <sup>(7)</sup>● Choledocholithiasis, actual or suspected and, **One:**● Pain, right upper abdomen and, ≥ **One:** <sup>(8, 9)</sup>– Common bile duct stone or dilation on imaging <sup>(10, 11)</sup>– Total bilirubin > 1.7 mg/dL and abnormal LFTs <sup>(12, 13)</sup>● **Episode Day 1, One:** <sup>(1, 3, 4)</sup>

(Symptom or finding within 24h)

● **OBSERVATION, One:** <sup>(5)</sup>● Acute cholecystitis and, **All:** <sup>(6)</sup>● Confirmed by imaging, ≥ **One:**

– Ultrasound (US)

– CT

– MRI

– Hepatobiliary scintigraphy

– Unresponsive to at least 2 doses of analgesics

– Analgesic 3x/24h or continuous

– Anti-infective

● **ACUTE, One:** <sup>(7)</sup>● Choledocholithiasis, actual or suspected and, **Both:**● Pain, right upper abdomen and, ≥ **One:** <sup>(8, 9)</sup>– Common bile duct stone or dilation on imaging <sup>(10, 11)</sup>– Total bilirubin > 1.7 mg/dL and abnormal LFTs <sup>(12, 13)</sup>● Intervention, ≥ **One:** <sup>(14)</sup>

– ERCP performed within 24h

– Endoscopic ultrasound performed within 24h

– MRCP performed within 24h

– NPO and IV fluid <sup>(15)</sup>

– Analgesic ≥ 3x/24h

● **Episode Day 2, One:** <sup>(4, 16, 17)</sup>● **OBSERVATION, One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** <sup>(18)</sup>

– Afebrile

– Tolerating PO <sup>(19)</sup>– Pain controlled or manageable <sup>(20)</sup>● **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**– Requiring cholecystectomy, open (**use General Surgical subset**)● Temperature requiring anti-infective and, **One:**– ≥ 100.4°F(38.0°C) <sup>(21)</sup>● Inadequate oral intake and, **Both:** <sup>(22)</sup>

– Advancing diet as tolerated

- IV fluid <sup>(15)</sup>
- **Pain uncontrolled and, Both:**
  - Pain assessment <sup>(23)</sup>
  - Requiring change in medication, dose, or frequency <sup>(24)</sup>
- **ACUTE, One:**
  - **Responder**, discharge expected today if clinically stable last 12h, **All:** <sup>(18)</sup>
    - Tolerating PO <sup>(19)</sup>
    - Laboratory values within acceptable limits <sup>(25)</sup>
    - Pain controlled or manageable
    - No complication or active comorbidity relevant to this episode of care <sup>(26)</sup>
  - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
    - **Cholelithiasis, actual or suspected and, ≥ One:** <sup>(14)</sup>
      - ERCP performed within 24h
      - Endoscopic ultrasound performed within 24h
      - Surgery performed within 24h <sup>(27)</sup>
      - NPO and IV fluid <sup>(15)</sup>
      - Analgesic ≥ 3x/24h
- **Episode Day 3, One:** <sup>(4, 28, 29)</sup>
  - **OBSERVATION, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** <sup>(18)</sup>
      - Afebrile
      - Tolerating PO <sup>(19)</sup>
      - Pain controlled or manageable <sup>(20)</sup>
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
      - For a condition other than acute cholecystitis use appropriate subset (Episode Day 1) based on clinical findings
      - Refer for secondary review
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** <sup>(18)</sup>
      - Tolerating PO <sup>(19)</sup>
      - Laboratory values within acceptable limits <sup>(25)</sup>
      - Pain controlled or manageable
      - No complication or active comorbidity relevant to this episode of care <sup>(26)</sup>
    - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
      - **Cholelithiasis and, ≥ One:** <sup>(14)</sup>
        - ERCP performed within 24h
        - Surgery performed within 24h <sup>(27)</sup>
      - **Inadequate oral intake and, ≥ One:** <sup>(22)</sup>
        - IV fluid <sup>(15)</sup>
        - Advancing diet as tolerated
      - Analgesic ≥ 3x/24h
  - **Episode Day 4, One:**
    - **ACUTE, One:**
      - **Responder**, discharge expected today if clinically stable last 12h, **All:** <sup>(18)</sup>
        - Tolerating PO <sup>(19)</sup>
        - Laboratory values within acceptable limits <sup>(25)</sup>
        - Pain controlled or manageable
        - No complication or active comorbidity relevant to this episode of care <sup>(26)</sup>
      - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
        - **Use Extended Stay subset**
        - Refer for secondary review

**Discharge, Level of Care, One:** <sup>(30)</sup>**● HOME, Both:****● Level of care appropriateness, Both:**

- Home environment safe and accessible <sup>(31)</sup>
- Patient or caregiver demonstrates ability to manage care

**● Acute cholecystitis or choledocholithiasis discharge planning, All:**

- Provide comprehensive written discharge and teaching instructions <sup>(32)</sup>
- Medication reconciliation <sup>(33)</sup>

**● Education on the importance of, All:**

- Notify physician immediately if fever, jaundice, or severe pain occurs
- Importance of maintaining adequate hydration
- Implementing healthy lifestyle modifications <sup>(34)</sup>
- Limiting activity <sup>(35)</sup>
- Follow-up care planned <sup>(36)</sup>
- Identify and address transportation needs

**● HOME CARE, Both:****● Level of care appropriateness, All:**

- Home environment safe and accessible <sup>(31)</sup>
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable <sup>(37, 38)</sup>

**● Skilled services, ≥ One:** <sup>(39)</sup>

- Skilled nursing <sup>(40)</sup>
- Skilled therapy, PT, OT, or ST
- Behavioral health <sup>(41)</sup>

**● Acute cholecystitis or choledocholithiasis discharge planning, All:**

- Provide comprehensive written discharge and teaching instructions <sup>(32)</sup>

**● Education on the importance of, All:**

- Notify physician immediately if fever, jaundice, or severe pain occurs
- Importance of maintaining adequate hydration
- Implementing healthy lifestyle modifications <sup>(34)</sup>
- Limiting activity <sup>(35)</sup>
- Medication reconciliation <sup>(33)</sup>
- Follow-up care planned <sup>(36)</sup>
- Identify and address transportation needs

**● BEHAVIORAL HEALTH, Both:****● Level of care appropriateness, One:**

- Support systems available <sup>(42)</sup>

**● Skilled treatment, ≥ One:**

- Clinical assessment <sup>(43)</sup>
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation <sup>(44)</sup>

**● SKILLED MEDICAL OR THERAPY, Both:****● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

**● Services, ≥ One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h <sup>(40)</sup>

**● Therapy, All:**

- Goal directed therapy with rehabilitation potential <sup>(45, 46)</sup>
- Functional impairments requiring at least supervision <sup>(47)</sup>
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

**● Complete prior to facility transfer, All:**

- Medication reconciliation <sup>(33)</sup>
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

● **Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight  $\geq 2x/wk$
- Treatment precluded in a lower level

● **Services,  $\geq$  One:**

- Medical and skilled nursing interventions or assessment 4h/24h <sup>(40)</sup>

● **Therapy, All:**

- Goal directed therapy with rehabilitation potential <sup>(45, 46)</sup>
- $\geq 2$  functional impairments requiring at least minimum assistance <sup>(47)</sup>
- Able to tolerate 2-3h/d of skilled therapy  $\geq 5d/wk$  and  $\geq 1$  therapy discipline

● **Complete prior to facility transfer, All:**

- Medication reconciliation <sup>(33)</sup>
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE REHAB, Both:**

● **Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight  $\geq 3x/wk$
- Treatment precluded in a lower level

● **Services, All:**

- Goal directed therapy with rehabilitation potential <sup>(45, 46)</sup>
- $\geq 2$  functional impairments requiring at least minimum assistance <sup>(47)</sup>
- Able to tolerate 2-3h/d of skilled therapy  $\geq 5d/wk$  and  $\geq 2$  therapy disciplines

● **Complete prior to facility transfer, All:**

- Medication reconciliation <sup>(33)</sup>
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● **Level of care appropriateness, Both:**

- Medical practitioner, NP, PA assessment or oversight  $\geq 3x/wk$

● **Services, All:**

- Goal directed therapy with rehabilitation potential <sup>(45, 46)</sup>
- $\geq 2$  functional impairments requiring at least minimum assistance <sup>(47)</sup>
- Able to tolerate  $\geq 15h/wk$  of skilled therapy and  $\geq 2$  therapy disciplines <sup>(48)</sup>

● **Complete prior to facility transfer, All:**

- Medication reconciliation <sup>(33)</sup>
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

● **Level of care appropriateness, Both:**

- Treatment precluded in a lower level

● **In lieu of acute or continued hospitalization or failed lower level of care,  $\geq$  One:**

- Primary condition or illness and treatment of active comorbid condition(s) <sup>(49)</sup>
- Ventilator dependent  $\geq 6h/d$  and weaning planned <sup>(50, 51)</sup>
- Acute and comorbid conditions requiring prolonged hospitalization <sup>(52)</sup>

● **Complete prior to facility transfer, All:**

- Medication reconciliation <sup>(33)</sup>

- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Utilization Benchmarks

| Condition or Procedure                            | % Pd Obs | LOS (days) | Type      |
|---------------------------------------------------|----------|------------|-----------|
| 444 DISORDERS OF THE BILIARY TRACT WITH MCC       | n/a      | 4.4        | CMS GMLOS |
| 445 DISORDERS OF THE BILIARY TRACT WITH CC        | n/a      | 3.1        | CMS GMLOS |
| 446 DISORDERS OF THE BILIARY TRACT WITHOUT CC/MCC | n/a      | 2.2        | CMS GMLOS |
| ACUTE CHOLECYSTITIS                               | 31%      | 3.3        | InterQual |
| CHOLEDOCHOLITHIASIS                               | 10%      | 3.2        | InterQual |

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

**Notes:****1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
  - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

**NOTE:** In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

**NOTE:** CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

**2:**

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

**Note:** While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and



intensity of service requirements.

*For further clarification, see the Acute Level of Care Review Process.*

**3:**

**Expected progress after 12-24h**

- Pain decreasing and manageable with medication
- Vital signs within acceptable limits
- PO intake improving or IV fluids infusing
- Imaging studies in progress or completed

**Consider:**

- Surgical evaluation
- Gastroenterologist evaluation

**4:**

**Care Facilitation**

Potential risk of readmission, consider an alternate level of care or readmission reduction program

- Knowledge deficit or non-adherence with condition management identified (**Home Care, Readmission reduction program**)

**Social Determinants of Health (SDOH)**

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

**Alcohol or substance use**

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

**Domestic violence and sexually abused individuals**

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

**End of life planning, arrange palliative care or hospice**

- Consider palliative care for patients with end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

**Financial resources strained or limited**

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, home homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

**Functional or ADL impairment**

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

**Mental health co-morbidities involved**

- Appointment with primary care provider or behavioral health specialist upon discharge

- Care coordination with outpatient behavioral health providers
- Community resources should be given

**Risk for readmission**

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

**Risk for treatment nonadherence**

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

**Environmental exposure to toxic substances and other physical hazards**

- For tobacco use and/or exposure information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies, i.e., blood lead level testing and follow up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

**Unstable living environment or homelessness expected upon discharge**

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

**Unsafe home environment**

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

**5:**

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

**Note:** InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

**6:**

In a hospitalized patient with fever, elevated white blood cells, and a positive Murphy's sign (abrupt termination of inspiration with palpation over gallbladder), ultrasound is the preferred imaging test for patients with suspected cholecystitis as it is (Gomes et al., World J Gastrointest Surg 2017, 9: 118-26; Knab et al., Surg Clin North Am 2014, 94: 455-70):

- Readily available
- Does not expose the patient to ionizing radiation
- Confirms the presence or absence of gallstones

- Allows for morphologic evaluation
- Allows for evaluation of the intra and extrahepatic bile ducts
- Facilitates the identification or exclusion of alternative diagnoses

**7:**

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

**Note:** InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

**8:**

**Instruction:** For this criteria point, pain should be reported as moderate to severe (e.g., 4 to 10) on a pain rating scale.

Pain associated with choledocholithiasis typically presents in the right upper quadrant but may also be reported in the epigastric region. Pain typically lasts more than 30 minutes and up to several hours and is not dependent on positioning (Cianci and Restini, World J Gastroenterol 2021, 27: 4536-54; European Association for the Study of the Liver . Electronic address, J Hepatol 2016, 65: 146-81).

**9:**

Visualization of common bile duct (CBD) stones on imaging, cholangitis, and a total bilirubin greater than 4 mg/dL with CBD dilation (greater than 6 mm with gallbladder in situ or greater than 8 mm if gallbladder was previously removed) are high-risk indicators for CBD stones. Individuals presenting with high-risk indicators have a greater than 50%(0.50) likelihood of having CBD stones (Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15). Proceeding with endoscopic retrograde cholangiopancreatography (ERCP) without further diagnostics is reasonable for these individuals and may eliminate the need for subsequent procedures and anesthesia (Narula et al., Surg Endosc 2020, 34: 1482-91; Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15; Manes et al., Endoscopy 2019, 51: 472-91). Individuals with intermediate predictors have a 10%(0.10) to 50%(0.50) likelihood of actually having CBD stones. Intermediate predictors include abnormal liver function tests, age greater than 55 years, or CBD dilation (greater than 6 mm with gallbladder in situ or greater than 8 mm if gallbladder was previously removed) (Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15). Because ERCP is associated with a 6%(0.06) to 15% (0.15) possibility of major adverse events (e.g., pancreatitis, bile duct injury), magnetic resonance cholangiopancreatography (MRCP), endoscopic ultrasound, or intraoperative cholangiogram are considered appropriate first-line investigative tools for individuals with intermediate predictors of CBD stones (Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15).

**10:**

Common bile duct dilation associated with choledocholithiasis is defined as greater than 6 mm with gallbladder in situ or greater than 8 mm if gallbladder has previously been removed.

**11:**

Common bile duct (CBD) dilation is associated with increased risk for choledocholithiasis. The American Society for Gastrointestinal Endoscopy defines CBD dilation as greater than 6 mm with gallbladder in situ or greater than 8 mm if gallbladder has previously been removed (Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15).

**12:**

For this criteria point, abnormal liver function tests include:

- Alanine aminotransferase (ALT) (serum glutamic pyruvic transaminase [SGPT])
- Aspartate aminotransferase (AST) (serum glutamic oxaloacetic transaminase [SGOT])
- Gamma-glutamyl transferase
- Alkaline phosphatase

**13:**

Elevated liver function tests (LFTs) are nonspecific markers for choledocholithiasis; however, they are recommended as part of a comprehensive initial evaluation (Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-

105.e15; Manes et al., Endoscopy 2019, 51: 472-91). Elevated LFTs (including 1 or more of total bilirubin, aspartate transaminase [AST], alanine transaminase [ALT], alkaline phosphatase, or gamma-glutamyl transferase) are present in most individuals with confirmed choledocholithiasis (Manes et al., Endoscopy 2019, 51: 472-91). In this case, elevated total bilirubin is defined as greater than 1.7 mg/dL (Narula et al., Surg Endosc 2020, 34: 1482-91). Because of the nonspecific nature of elevated LFTs, further diagnostics are warranted to confirm the diagnosis of choledocholithiasis before a treatment plan is decided (Kadah et al., Hepatobiliary Pancreat Dis Int 2020, 19: 266-70; Narula et al., Surg Endosc 2020, 34: 1482-91; Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15; Manes et al., Endoscopy 2019, 51: 472-91).

**14:**

For individuals with symptomatic choledocholithiasis, common bile duct (CBD) stone extraction is recommended. Endoscopic retrograde cholangiopancreatography (ERCP) is the gold standard treatment for the extraction of CBD stones (Narula et al., Surg Endosc 2020, 34: 1482-91; Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15). Proceeding directly to ERCP for CBD stone extraction is reasonable when there is a confirmed diagnosis of choledocholithiasis; however, further diagnostics with intraoperative cholangiogram, endoscopic ultrasound, or noninvasive imaging (e.g., computed tomography [CT] scan, magnetic resonance cholangiopancreatography [MRCP]) is recommended if there is not a confirmed diagnosis (Narula et al., Surg Endosc 2020, 34: 1482-91; Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15). When CBD stones are confirmed, the timing (preoperative, intraoperative, or postoperative) of stone extraction by ERCP or CBD exploration is a matter of facility resources and specialist availability (Boni et al., Surg Endosc 2022, 36: 7863-76; Narula et al., Surg Endosc 2020, 34: 1482-91; Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15; Manes et al., Endoscopy 2019, 51: 472-91). Although a systematic review and meta-analysis suggest that performing intra-operative ERCP or laparoscopic CBD exploration is favorable when compared to preoperative or postoperative intervention, authors acknowledge that there is a lack of strong evidence available to make a recommendation for the proposed timing and that pre-operative ERCP is the most prevalent practice (performed greater than 80% of the time) due to facility resources and specialist availability (Boni et al., Surg Endosc 2022, 36: 7863-76).

**15:**

**Instruction:** This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

**16:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights. For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

**NOTE:** For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

**17:****Expected progress after 48h**

- Pain decreasing and manageable with medication
- Afebrile with stable vital signs and within acceptable limits
- Adequate oral nutrition or IV fluids infusing
- Imaging studies in progress or completed
- Laboratory values normalizing

**If not, consider:**

- Surgical consult
- Gastroenterology consult

**18:**

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

**19:**

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

**20:**

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

**21:**

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

**22:**

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).



**23:**

Pain assessment includes quality, intensity, location, frequency, duration, and exacerbating or alleviating factors. Uncontrolled pain should be monitored, assessed, and documented. Appropriate medication adjustment and other interventions should be taken to reduce pain.

**24:**

**Instruction:** For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

**25:**

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

**26:**

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

**27:**

**Instruction:** This criterion is for patients who have had an outpatient surgery (e.g., laparoscopic cholecystectomy, intraoperative cholangiogram, common bile duct exploration) performed within the previous 24 hours. For inpatient surgical procedures (e.g., open cholecystectomy) see the General Surgical subset.

Individuals with suspected or confirmed choledocholithiasis may require surgical intervention that can be performed either concurrently with a cholecystectomy or independent from a cholecystectomy. Surgical procedures include intraoperative cholangiogram, common bile duct exploration, or the placement of a biliary stent to facilitate biliary drainage (Boni et al., Surg Endosc 2022, 36: 7863-76; Buxbaum et al., Gastrointest Endosc 2019, 89: 1075-105.e15; Manes et al., Endoscopy 2019, 51: 472-91).

**28:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

**Note:** Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

**29:****Expected progress within 72h**

- Pain manageable with or without medication
- Oral nutrition adequate or improving
- Laboratory values normalizing or within acceptable limits

**Consider:**

- Discharge planning

**30:**

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

**31:**

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

**32:**

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

**33:**

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

**34:**

Symptomatic gallbladder disease is associated with lifestyle choices, and incidences of acute inflammation and development of bile stones may be reduced by implementing healthy lifestyle changes (Wirth et al., Am J Clin Nutr 2020, 112: 586-94). Individuals with symptomatic gallbladder disease should be encouraged to make healthy lifestyle changes, such as maintaining a healthy weight, limiting consumption of alcohol and caffeine, eating a healthy diet, and engaging in regular physical activity (European Association for the Study of the Liver . Electronic address, J Hepatol 2016, 65: 146-81).

**35:**

Limitations on activity will vary based on individualized patient care and interventions. Individuals who received surgical interventions (e.g., cholecystectomy) may require longer activity restrictions than individuals who received noninvasive medical care (e.g., anti-infectives) or less invasive procedural interventions (e.g., endoscopic retrograde cholangiopancreatography [ERCP]).

**36:**

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

**37:**

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

**38:**

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

**39:**

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

**40:**

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

**41:**

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care.



When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

#### **Evaluation and Treatment**

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

#### **42:**

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

#### **43:**

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

#### **44:**

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

#### **45:**

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)

- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

**46:**

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

**47:**

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

**48:**

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

**49:**

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

**50:**

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

**51:**

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

**52:**

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.